**Design Journal Summary**
FEEG6013 Group Design Project

# 33
# Reconfigurable Origami Antenna; Mechanism of Deployment for Spacecraft structures, ROAM-DS

Developing reconfigurable deployable space communication systems


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# 1. Summary of Individual Contribution & Key Achievements

ROAM-DS was a novel proof of concept project which made use of origami structures in the context of a spacecraft communication system to alter the communication parameters of said communication system. The project made use of second order derivatives of a Miura-ori origami pattern which had been previously developed by the supervisors Dr Naskar and Dr Mukhopadhyay as well as a PhD student Aditi Sharma to alter the curvature of a parabolic communication disk, making use of the compactness such origami patterns allow in their folded state for space applications as well as allowing for a reconfiguration of the curvature of the parabolic dish throughout the folding of the structure.

In this project, I was on the computational modelling team with my core reasonability being to adapt the aforementioned second order derivatives of the Miura-ori origami from thin panel origami to thick panel origami. My core deliverable was to develop a computational model of the final origami pattern chosen for the communication dish as well as to develop computational models for all of the second order Miura-ori patterns that had been developed by Aditi. Throughout this project, I researched various methods for strategies for creating thick panel origami structures as well as performing independent derivations to convert the thin panel geometries into thick panel ones. By leveraging computational tools I was able to create a model for the base Miura-ori pattern which modelling the folding behavior of the thick panel geometry as well as outlining a strategy that can be expanded to model more complex geometries.

# 2. Technical Contribution

Due to the novelty involved in developing a computational model of thick panel Miura-ori geometries, the development process was decomposed into discrete stages. Each stage would build on the previous one and contribute to the final model. The first stage consisted on analysis of relevent literature as well as past work on the Miura-ori geometries as well as on thick panel origami. The next stage was to derive any relevant equations or relationships that would be utilized within the program. The final stage was to develop the program itself, bringing together equations derived in the previous section to create a model.

## 2.1. Preliminary Design Research

For this project, it was critical to develop an understanding of the Miura-ori geometry as well as its derivatives and what advantageous properties this this family of origami geometries posses. As my focus within the project was on adapting thin panel Miura-ori architectures for thick panel as well as computationally modelling these thick panel variants, further focus was applied on novel thick panel geometry sas well as the computational methods used to simulate origami structures. Motivated by this, a comprehensive literature review was undertaken with the following goals:

1. Understand the kinematic and geometrical derivations needed to model thin panel origami structures and adapt these to thick panel origami structures.
2. Ascertain the benefits of thick panel origami over thin panel specifically within the constant of Miura-ori architectures and its derivatives.
3. Note novel thick panel geometries which can be used for the Miura-ori family of origami structures.
4. Identify the computational methods used for modelling the kinematic behaviour (folding behaviour) of origami structures.

Origami itself is defined as the art of folding objects out of paper to create both two-dimensional and three-dimensional subjects **[1]** with its origins dating back to ancient Japan. In the modern day, the papercraft principles of origami have seen widespread use in various disciplines such as architecture, robotics and spacecraft structures.Whereas origami uses only folds, kirigami is a closely related subset which uses folds and cuts to create more complex structures.

Metamaterials is a novel branch of material science which encompass three dimensional structures with a response, be that electromagnetic, acoustic, mechanical or thermal, due to the collective effect of meta-atom elements that is not achieve conventionally with any individual constituent material **[2]**. The response of metamaterials does not originate from the chemical composition of the constituent materials but from teh structure of the meta-atom itself. A metasurface is a subset of metamaterials where the structural elements are confined to a two dimensional plane instead of a three dimensional structure **[2]**. Some advantageous mechanical properties of engineered metamaterials are negative Poisson's ratio (Auxeticity), shape morphing and reconfigurability, multistability and high specific stiffness or strength. Many metamaterials make use of origami and or kirigami meta-atom architectures to achieve such mechanical properties.

The Miura-ori origami pattern is a fundamental origami structure which sees four intersecting crease lines which are symmetrical about a horizontal center line **[3]**, the base Miura-ori architecture is shown in **Figure 1**.

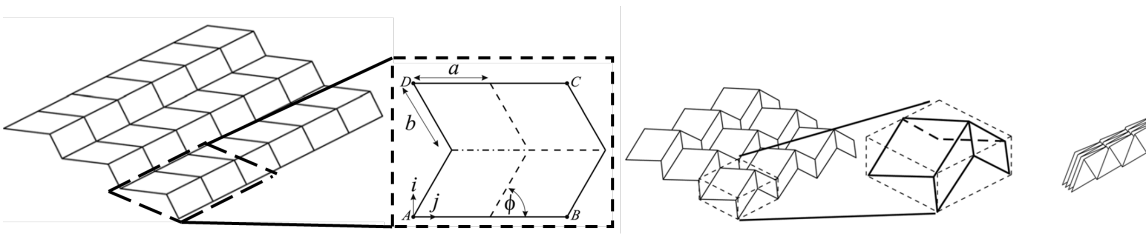


Figure 1: Image depicting the base Miura-ori architecture in an unfolded state with annotated repeated unit [Left], a partially folded state [Center] and a fully folded state [Right] **[3]**.

The geometry itself is constructed with repeating parallelogram facets which themselves are defined by the side lengths $a$ and $b$ and the sector angle $\phi$. The structure as a whole exhibits both auxetic and reconfigurable behaviour arising from the crease configuration of the panels. Some other useful properties identified by the literature **[3]** are:

- Developability: The Miura-ori architecture can be folded from a continuous flat sheet;
- Flat-Foldability: All panels are coplanar when the pattern is fully folded.
- Tessellation: The Miura-ori architecture utilizes a repetitive unit cell geometry constructed from a single repeating plate size.
- Single DoF motion: The Miura-ori architecture possesses a single Degree-of-Freedom (DoF) throughout its motion, irrespective of the size of the array **[4]**.

These properties have caused the Miura-ori architecture to see widespread use in various fields as such as in solar panels for the Space Flyer Unit satellite **[4]**, in increasing the impact resistance of structures **[5]**, in deployable antennas **[6]** and in reconfigurable electromagnetic metamaterials for tunable wave manipulation **[7]**.

Further advantageous properties can be obtained by altering a characteristic of the base Miura-ori geometry, such modified architectures are called first level derivatives. By changing the crease alignment of the zig-zag lines present between adjacent unit cells of the in the base Miura geometry, defines the Arc-Miura geometry, shown in **Figure 2**. If instead the rectilinearity of the base miura unit cell is changed, by changing the angles of the straight lines defining each panel so that they converge, the tapered Miura is defined shown in **Figure 3**.

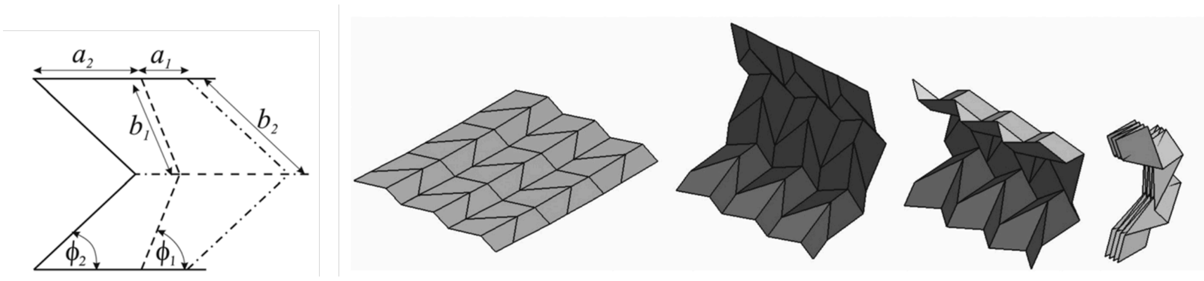

Figure 2: Image depicting a unit cell of the Arc-Miura architecture [Left] and the folding behaviour of the tessellated structure [Right] **[3]**.

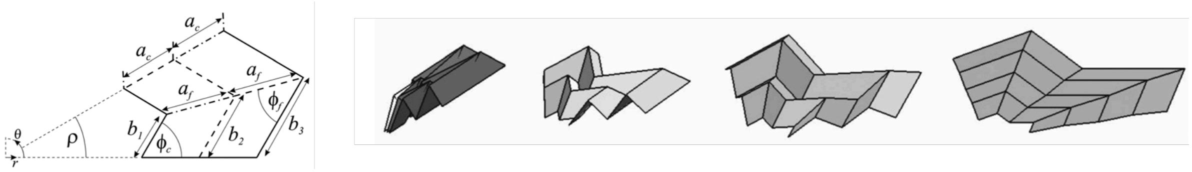

Figure 3: Image depicting a unit cell of the tapered Miura architecture [Left] and the folding behaviour of the tessellated structure [Right] **[3]**.

By altering two or more characteristics of the base Miura-ori geometry, second level derivatives can be defined. By altering both the crease alignment and by inclining each straight line by a constant angle, the inclined Arc-Miura geometry is defined and acts as an example of a second level derivative of the base geometry **[8]**. This architecture is shown in **Figure 4**.

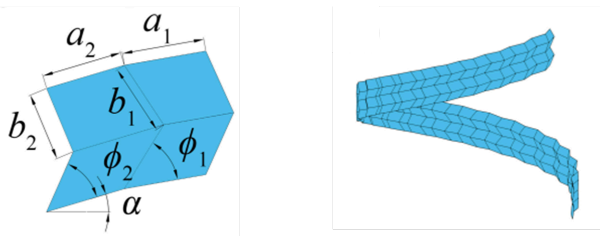

Figure 4: Image depicting a unit cell of the inclined Arc-Miura architecture [Left] and the folding behaviour of the tessellated structure [Right] **[8]**.

Thin panel origami has been the focus of much of the analysis and simulations on origami structures however as the thickness of the panels increase, the effect of the thickness of the panel on the kinematics of the structure cannot be neglected. Furthermore, thin panel origami simplifies crease behaviour and further neglects geometry intersection. Therefore, thick panel origami structures aim to encompass panel thickness within the kinematic behaviour **[9]** allowing for more realistic representation of the folding behaviors of origami structures. However, the analysis of thick panel origami necessitates more complex modelling approaches and generating the geometry of such origami structures also necessitates more complex rulesets.

Many methods exist to computationally model the kinematic behaviour of origami structures **[10]** such as utilizing basic trigonometric relationships to compute nodal positions for thin origami structures to utilizing spatial linkages to model the kinematic behaviour of thick panel structures.

One gap in the literature was combining many of the aforementioned concepts together. Mainly taking first and second level derivatives of the base Miura-ori architecture and expanding their kinematic equations for the thin panel configurations of these geometries to thick panels. A computational model of these thick panel structures could leverage both trigonometric equations and spatial linkages to allow for full definition of the structure throughout its folding, allowing for a tool to simulate metasurfaces made from meta-atoms of Miura-ori derivates.

## 2.2. Aims and Objectives For Computational Modelling

The aim of the computational modelling work within ROAM-DS was to alter the existing kinematic relationships for thin panel Miura-ori architectures to thick panel variants and then implement these relationship within a simulation to model the three dimensional folding behaviour of such structures. The intended long term goal of this simulation was to model the behaviour of higher level derivatives specifically the final thick-panel derivative architecture selected for the ROAM-DS communication dish. The decided objectives set to achieve this aim are as follows:

- To adapt the existing kinematic relationships for thin panel Miura-ori origami architectures to thick panel.
- To identify an expandable thick panel geometry that can be simulated.
- To create a ruleset to generate cartesian coordinates for the thick panel variant of the base Miura-ori geometry.
- To design a simulation architecture for the thick panel variant of the base Miura-ori geometry.
- To expand the ruleset and simulation to implement thick panel variants of first level derivates of the Miura-ori architecture.
- To combine the rulesets of multiple first level derivatives to yield second level derivatives of the Miura-ori geometry.
- To implement the final thick panel higher level derivative architecture chosen for thee ROAM-DS communication dish within the completed simulation framework.

## 2.3. Kinematic Analysis of Thick Panel Origami Structures

As was identified within the literature, one common method for simulating the kinematics of origami structures is through utilizing spatial linkages. In this method, the origami structure is assumed to be rigid in that the panels are treated as non-deformable bodies with movement restricted to one DoF rotation about crease lines. Spatial linkages model these creases specifically as revolute joints and the panels themselves and rigid linkages **[11]**.

The conversion between the coordinate system of one revolute joint to the next is done through the definition of Denavit-Hartenberg (DH) parameters within a transformation matrix, a general configuration of joints is shown in **Figure 5**. The four DH parameters are defined below:

- $\theta$: Angle about Z axis of the joint ($Z_i$), measured from the X axis of previous joint ($X_i$) to the X axis of the next joint ($X_{i+1}$).
- $a$: Length along the X axis of the joint ($X_i$) from joint $i$ to joint $i+1$.
- $R$: Offset along the Z axis of the joint ($Z_i$) from joint $i$ to joint $i+1$.
- $\beta$: Angle about X axis of the joint ($X_i$), measured from the Z axis of previous joint ($Z_i$) to the Z axis of the next joint ($Z_{i+1}$).

Unlike the general configuration shown in **Figure 5**, in the Miura-ori architecture the four revolute joints modelling the four intersecting creases form a close loop, allowing for the transformation matrices to be linked together within a closure condition. This closure condition generates a system of equations which can be simplified to yield equations describing the kinematics of the linkage in terms of the four DH parameters.

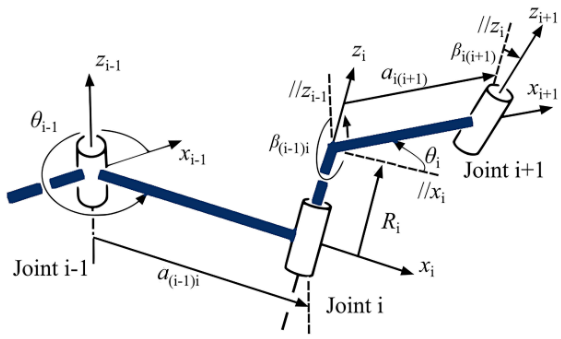


Figure 5: Image depicting the four Denavit-Hartenberg parameters ($\theta, a, R, \beta$). **[12]**

For thin panel configurations of the Miura-ori architecture, the four creases intersect at a point in the center of the unit cell whereas for thick panel configuration the thickness of each panel prevents this. A diagram depicting this is shown in **Figure 6**.

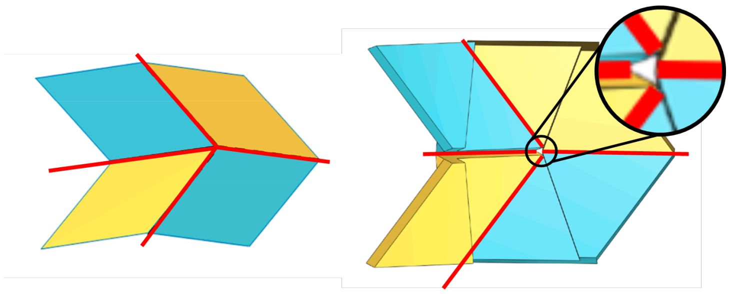

Figure 6: Image depicting the crease lines (shown in red) on a thin panel Miura-ori unit cell meeting at one point [Left] and crease lines not intersecting at one point for a thick panel configuration of the Miura-ori unit cell [Left.] **[13]**

This results in a differing set of simplifying conditions being imposed on the set of DH matrices for each panel configuration, this further yields a different configuration of spatial linkages.For thin panel origami a spherical linkage can be utilized to model the system of joints as both the $a$ and $R$ parameters are zero, this linkage is shown in **Figure 7**. For thick panel configurations a Bennett linkage must be used, as $R = 0$ but $a \neq 0$. This linkage is shown in **Figure 8**.

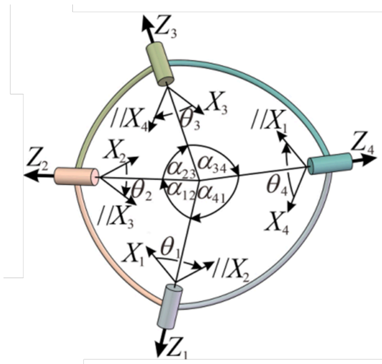


Figure 7: Image depicting the spherical linkage used to simplify the kinematic behaviour of thin panel Miura-ori configurations ($\alpha = \beta$). **[11]**

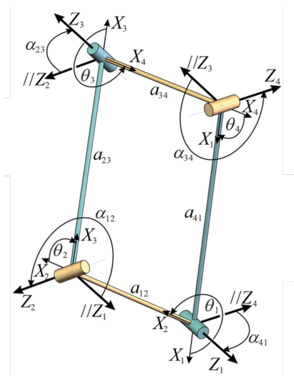

Figure 8: Image depicting the Bennett linkage used to simplify the kinematic behaviour of thick panel Miura-ori configurations ($\alpha = \beta$). **[11]**

The spherical linkage configuration for thin panel origami meant that the $4 \times 4$ DH transformation matrix for each joint could be reduced to $3 \times 3$, resulting in nine kinematic equations. As $a \neq 0$ this reduction in matrix size cannot be done for thick panel configurations, and so 16 equations are generated of which three contain $a$ terms. These set of equations can be simplified by imposing geometrical constraints derived from the Miura-ori geometry, these are shown in **Eq. 1** where $\beta$ simplifications can be done for both thin and thick panel configuration but $a$ simplifications can only be done for thick panel configurations.

$$\beta_{12} = \pi - \beta_{34} \quad \beta_{23} = \pi - \beta_{41} \quad a_{12} = a_{34} \quad a_{23} = a_{41} \quad \frac{a_{12}}{a_{23}} = \frac{\sin(\beta_{12})}{\sin(\beta_{34})} \tag{1}$$

Applying the conditions in **Eq. 1** to the set of thick panel equations, it was found that the three equations with $a$ terms appeared as either a common factor on both sides of the equation or formed an expression equal to zero. As a result the $a$ terms could be cancelled out of the closure condition equations. The complete derivation depicting this is contained within **Appendix A**. Therefore, under the geometrical assumptions contained within **Eq. 1**, the panel thickness did change the configuration of the spatial linkages but had no impact on the kinematic equations. The remaining non-trivial equations simplify in the same manner as the thin origami configuration resulting in the set of equations shown in **Eq. 2** **[12]** which can be used to model the kinematic behaviour of thin or thick panel origami.

$$\frac{\tan\theta_3}{\tan\theta_2} = -\frac{\cos\frac{\beta_{23}-\beta_{12}}{2}}{\cos\frac{\beta_{23}+\beta_{12}}{2}} \qquad \frac{\tan\theta_3}{\tan\theta_4} = \frac{\cos\frac{\beta_{23}-\beta_{12}}{2}}{\cos\frac{\beta_{23}+\beta_{12}}{2}} \tag{2.1}$$

$$\frac{\tan\theta_1}{\tan\theta_2} = -\frac{\cos\frac{\beta_{23}-\beta_{12}}{2}}{\cos\frac{\beta_{23}+\beta_{12}}{2}} \qquad \frac{\tan\theta_4}{\tan\theta_1} = \frac{\cos\frac{\beta_{23}-\beta_{12}}{2}}{\cos\frac{\beta_{23}+\beta_{12}}{2}} \tag{2.2}$$

## 2.4. Relating Kinematic and Geometric Parameters

The expressions in **Eq. 2** describe kinematic variables (DH parameters) however, these parameters must be linked to geometric parameters related to the unit cell geometry itself. Kinematically, the $\beta$ parameter defines the angle between consecutive Z axis, but geometrically this can be linked to the sector angle $\phi$ for both thick and thin panels. This is shown in **Figure 9** and **Figure 10**

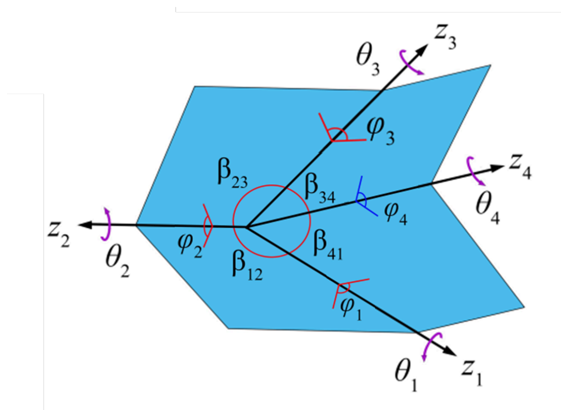


Figure 9: Image depicting the kinematic variables $\theta$ amd $\beta$, the dihedral angle $\varphi$ and the joint $z$ axis labelled on a thin panel Miura-ori unit cell. **[12]**.

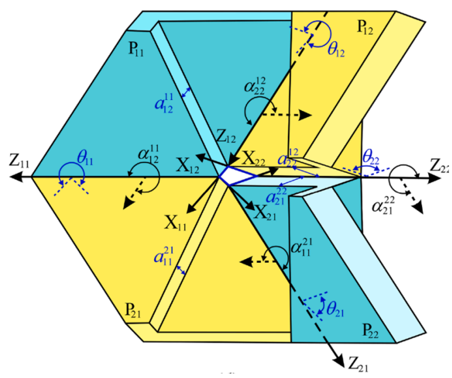


Figure 10: Image depicting the kinematic variables and joint axes labelled on a thick panel Miura-ori unit cell.**[13]**. The coordinate system is different from the one assumed in this report.

Utilizing both **Figure 9** and **Figure 10** the $\beta$ parameter can be directly related to the sector angle $\phi$, these relationships further govern the flat foldability of Miura-ori architectures **[14]**. The relationship between these parameters are shown in **Eq. 3**.

$$\beta_{12} = \phi \qquad \beta_{23} = \phi \tag{3}$$

The remaining kinematic parameter $\theta$ can be transformed into the dihedral angle ($\varphi$) which is a geometrical parameter defined as the angle between two panel faces about a crease line (shown in **Figure 9**). Despite the panel thickness having no impact on the kinematics, it does however impact the relationship between $\theta$ and $\varphi$. The thin panel and thick panel variants of this relationship are shown in **Eq. 4** respectively **Eq. 5**.

Thin panel relationships

$$\theta_1 = \varphi_1 + \pi \qquad \theta_2 = \pi - \varphi_2 \tag{4.1}$$
$$\theta_3 = \pi - \varphi_3 \qquad \theta_4 = \pi - \varphi_4 \tag{4.2}$$

Thick panel relationships

$$\theta_1 = \varphi_1 \qquad \theta_2 = \pi - \varphi_2 \tag{5.1}$$
$$\theta_3 = 2\pi - \varphi_3 \qquad \theta_4 = \pi + \varphi_4 \tag{5.2}$$

## 2.5. Derivation for the Edge Angles for Thick Panel Origami.

Gattas defines a geometrical parameter called the edge angle which can be used to tessilate the Miura-ori unit cell. Gattas defines the equations for these in [equation link] and shown in [image link].

It was important to validate that this edge angkle is not itself impacted by the thickness and so the equations had to be redireved.

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## Appendix A: Thick Origami Kinematic Derivation

Each crease in the Miura-Ori geometry can be modelled as a revolute hinge, with Denavit-Hartenberg (DH) matrices allowing for transfromation between the coordinate system of one hinge to the other. A general DH matrix as well as an inverse DH matrix are shown in **Eq. 6** and **Eq. 7** respectively.

$$T_{(i+1)i} = \begin{bmatrix} \cos\theta_i & -\cos\beta_{i(i+1)}\sin\theta_i & \sin\beta_{i(i+1)}\sin\theta_i & a_{i(i+1)}\cos\theta_i \\ \sin\theta_i & \cos\beta_{i(i+1)}\sin\theta_i & -\sin\beta_{i(i+1)}\sin\theta_i & a_{i(i+1)}\sin\theta_i \\ 0 & \sin\beta_{i(i+1)} & \cos\beta_{i(i+1)} & R_i \\ 0 & 0 & 0 & 1 \end{bmatrix}_{4\times4} \tag{6}$$

$$T_{i(i+1)} = T^{-1}_{(i+1)i} = \begin{bmatrix} \cos\theta_i & \sin\theta_i & 0 & -a_{i(i+1)} \\ -\cos\beta_{i(i+1)}\sin\theta_i & \cos\beta_{i(i+1)}\cos\theta_i & \sin\beta_{i(i+1)} & -R_i\sin\beta_{i(i+1)} \\ \sin\beta_{i(i+1)}\sin\theta_i & -\sin\beta_{i(i+1)}\cos\theta_i & \cos\beta_{i(i+1)} & -R_i\cos\beta_{i(i+1)} \\ 0 & 0 & 0 & 1 \end{bmatrix}_{4\times4} \tag{7}$$

For thin panel origami architectures $R = 0$ and $a = 0$ allowing for **Eq. 6** and **Eq. 7** to be reduced to $3 \times 3$ matrices. For thick panel origami however, $a \neq 0$ and the resulting linkage is a bennet linkage. The closure condition however remains the same as the thin panel configuration and is shown in **Eq. 8**.

$$[T_{2\to1}]_{4\times4}[T_{3\to2}]_{4\times4}[T_{4\to3}]_{4\times4} = [T_{4\to1}]_{4\times4} \tag{8}$$

The transformation matrices can be simplified further by applying the constraints of the Miura-Ori geometry shown in **Eq. 1**. Utilizing these conditions, the simplifying substitutions shown in **Eq. 9** will be made into the transformation matrices.

$$\beta_{34} = \pi - \beta_{12} \quad \beta_{41} = \pi - \beta_{23} \tag{9.1}$$

$$a_{34} = a_{12} = a_{23}\frac{\sin(\beta_{12})}{\sin(\beta_{23})} = a_{41}\frac{\sin(\beta_{12})}{\sin(\beta_{23})} \tag{9.2}$$

The resulting transformation matrices for each revolute hinge is shown in **Eq. 10**, **Eq. 11**, **Eq. 12** and **Eq. 13**, with **Eq. 13** being the inverse matrix utilizing the form shown in **Eq. 7**.

$$[T_{2\to1}]_{4\times4} = \begin{bmatrix} \cos(\theta_1) & -\cos(\beta_{12})\sin(\theta_1) & \sin(\beta_{12})\sin(\theta_1) & \frac{a_{23}\sin(\beta_{12})\cos(\theta_1)}{\csc(\beta_{23})} \\ \sin(\theta_1) & \cos(\beta_{12})\cos(\theta_1) & -\sin(\beta_{12})\cos(\theta_1) & \frac{a_{23}\sin(\beta_{12})\sin(\theta_1)}{\csc(\beta_{23})} \\ 0 & \sin(\beta_{12}) & \cos(\beta_{12}) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}_{4\times4} \tag{10}$$

$$[T_{3\to2}]_{4\times4} = \begin{bmatrix} \cos(\theta_2) & -\cos(\beta_{23})\sin(\theta_2) & \sin(\beta_{23})\sin(\theta_2) & a_{23}\cos(\theta_2) \\ \sin(\theta_2) & \cos(\beta_{23})\cos(\theta_2) & -\sin(\beta_{23})\cos(\theta_2) & a_{23}\sin(\theta_2) \\ 0 & \sin(\beta_{23}) & \cos(\beta_{23}) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}_{4\times4} \tag{11}$$

$$[T_{4\to3}]_{4\times4} = \begin{bmatrix} \cos(\theta_3) & \cos(\beta_{12})\sin(\theta_3) & \sin(\beta_{12})\sin(\theta_3) & \frac{a_{23}\sin(\beta_{12})\cos(\theta_3)}{\sin(\beta_{23})} \\ \sin(\theta_3) & -\cos(\beta_{12})\cos(\theta_3) & -\sin(\beta_{12})\cos(\theta_3) & \frac{a_{23}\sin(\beta_{12})\sin(\theta_3)}{\sin(\beta_{23})} \\ 0 & \sin(\beta_{12}) & -\cos(\beta_{12}) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}_{4\times4} \tag{12}$$

$$[T_{4\to1}]_{4\times4} = \begin{bmatrix} \cos(\theta_4) & \sin(\theta_4) & 0 & -a_{23} \\ \cos(\beta_{23})\sin(\theta_4) & -\cos(\beta_{23})\cos(\theta_4) & \sin(\beta_{23}) & 0 \\ \sin(\beta_{23})\sin(\theta_4) & -\sin(\beta_{23})\cos(\theta_4) & -\cos(\beta_{23}) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}_{4\times4} \tag{13}$$

Substituting these four transformation matrices (**Eq. 10**, **Eq. 11**, **Eq. 12** and **Eq. 13**,) into the closure condition (**Eq. 8**) yields 16 equations. These 16 equations can be assigned an index depending on which parameter in each matrix the equation links to, for example Eq (4,2) corresponds to the equation generated by the terms in the 4th row and second column of each matrix.

Using this index, equations Eq (4,1), Eq (4,2), Eq (4,3) and Eq (4,4) are all trivial equations (either $0 = 0$ or $1 = 1$). The equations generated by the top left nine terms of each matrix Eq (1,1) to Eq (3,3) match the equations generated by the spherical linkage for thin panel origami architectures. The remaining three equations (Eq (1,4), Eq (2,4) and Eq (3,4)) contain thickness terms ($a$) within them, these equations are shown in **Eq. 14**, **Eq. 15** and **Eq. 16**.

Eq (1,4)

$$A = \cos(\theta_1)\cos(\theta_2) - \cos(\beta_{12})\sin(\theta_1)\sin(\theta_2) \tag{14.1}$$

$$B = -\cos(\beta_{12})\cos(\beta_{23})\sin(\theta_1)\cos(\theta_2)... \tag{14.2}$$

$$... + \sin(\beta_{12})\sin(\beta_{23})\sin(\theta_1) - \cos(\beta_{23})\cos(\theta_1)\sin(\theta_2) \tag{14.3}$$

$$\tag{14.4}$$

$$a_{23}\left(\frac{A\sin(\beta_{12})\cos(\theta_3)}{\sin(\beta_{23})} + \frac{B\sin(\beta_{12})\sin(\theta_3)}{\sin(\beta_{23})} + \frac{\sin(\beta_{12})\cos(\theta_1)}{\sin(\beta_{23})} + A\right) = -a_{23} \tag{14.5}$$

Eq (2,4)

$$C = \cos(\theta_2)\sin(\theta_1) + \cos(\beta_{12})\cos(\theta_1)\sin(\theta_2) \tag{15.1}$$

$$D = \cos(\beta_{12})\cos(\beta_{23})\cos(\theta_1)\cos(\theta_2)... \tag{15.2}$$

$$... - \cos(\theta_1)\sin(\beta_{12})\sin(\beta_{23}) - \cos(\beta_{23})\sin(\theta_1)\sin(\theta_2) \tag{15.3}$$

$$\tag{15.4}$$

$$a_{23}\left(\frac{C\sin(\beta_{12})\cos(\theta_3)}{\sin(\beta_{23})} + \frac{D\sin(\beta_{12})\sin(\theta_3)}{\sin(\beta_{23})} + \frac{\sin(\beta_{12})\sin(\theta_1)}{\sin(\beta_{23})} + C\right) = 0 \tag{15.5}$$

Eq (3,4)

$$E = \sin(\beta_{12})\cos(\beta_{23})\cos(\theta_2) + \cos(\beta_{12})\sin(\beta_{23}) \tag{16.1}$$

$$\tag{16.2}$$

$$a_{23}\left(\frac{\sin^2(\beta_{12})\sin(\theta_2)\cos(\theta_3)}{\sin(\beta_{23})} + \frac{E\sin(\beta_{12})\sin(\theta_3)}{\sin(\beta_{23})} + \sin(\beta_{12})\sin(\theta_2)\right) = 0 \tag{16.3}$$

In these three equations an $a_{23}$ term can be factored out, and given that $a_{23} \neq 0$ which is true for thick panel origami it can be removed from each equation. This implies that panel thickness has no impact on the kinematics of the origami structure. The remaining nine non-trivial equations (Eq (1,1) to Eq (3,3)) can then be manipulated using trigonometric identities **[12]** to yield the four kinematic relationships shown in **Eq. 2**.